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METHODS AND APPARATUS TO CONNECT A SEMICONDUCTOR PACKAGE TO A CIRCUIT BOARD

Abstract

Systems, apparatus, articles of manufacture, and methods to connect a semiconductor package to a circuit board are disclosed. An example apparatus includes an interposer having a first side and a second side opposite the first side. A semiconductor package couples to the first side of the interposer via first contacts on a first surface of the semiconductor package. The semiconductor package includes second contacts on the first surface of the semiconductor package. The second contacts spaced apart from the first contacts. The second contacts to electrically couple to third contacts in a socket dimensioned to receive the interposer and the semiconductor package.

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Background/Summary

FIELD OF THE DISCLOSURE

[0001] This disclosure relates generally to semiconductor packages and, more particularly, to methods and apparatus to connect a semiconductor package to a circuit board.

BACKGROUND

[0002] In recent years, the variety of electronic devices which incorporate semiconductor or integrated circuit (IC) packages have grown considerably. Some such electronic devices include cellular phones, portable computers, hand-held devices, etc. These electronic devices typically include a circuit board on which a number of semiconductor packages are secured to provide multiple electronic functions. Different approaches to connecting integrated circuits to underlying circuit boards includes using monolithic packages and using package-on-interposer (PoINT) technology.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. **1** is a cross-sectional view of an example integrated circuit (IC) device assembly including an example package-on-interposer (PoINT) device inserted into an example socket on a circuit board in accordance with teachings disclosed herein.

[0004] FIG. **2** is a cross-sectional view of the example IC device assembly of FIG. **1** with the example PoINT device spaced apart from example socket.

[0005] FIG. **3** is an exploded view of the example IC device assembly of FIG. **1**.

[0006] FIG. **4** is a bottom view of an example semiconductor package of the example PoINT device of FIGS. **1-3**.

[0007] FIG. **5** is a cross-sectional view of another example IC device assembly in accordance with teachings disclosed herein.

[0008] FIG. **6** is a cross-sectional view of another example IC device assembly in accordance with teachings disclosed herein.

[0009] FIG. **7** is a flowchart representative of an example method to manufacture any one of the example IC device assemblies of FIGS. **1-6**.

[0010] FIG. **8** is a top view of a wafer including dies that may be included in an IC package constructed in accordance with teachings disclosed herein.

[0011] FIG. **9** is a cross-sectional side view of an IC device that may be included in an IC package constructed in accordance with teachings disclosed herein.

[0012] FIG. **10** is a cross-sectional side view of an IC package constructed in accordance with teachings disclosed herein.

[0013] FIG. **11** is a cross-sectional side view of an IC device assembly that may include an IC package constructed in accordance with teachings disclosed herein.

[0014] FIG. **12** is a block diagram of an example electrical device that may include an IC package constructed in accordance with teachings disclosed herein.

[0015] In general, the same reference numbers will be used throughout the drawing(s) and accompanying written description to refer to the same or like parts. The figures are not necessarily to scale. Instead, the thickness of the layers or regions may be enlarged in the drawings. Although the figures show layers and regions with clean lines and boundaries, some or all of these lines and/or boundaries may be idealized. In reality, the boundaries and/or lines may be unobservable, blended, and/or irregular.

DETAILED DESCRIPTION

[0016] Package-on-Interposer (PoINT) technology is a cost-effective solution helping to realize smaller packages relative to monolithic packages. As a result, PoINT technology can significantly lower package costs relative to monolithic packages. However, the performance of PoINT devices degrades compared to monolithic packages because there are more physical transition layers. Specifically, PoINT devices include one or more semiconductor dies and/or packages (sometimes also referred to as patches) mounted onto an interposer that may be used to interconnect the dies and/or packages with a circuit board (e.g., through a socket constructed to receive the interposer). Thus, circuit assemblies using PoINT devices include an additional interconnect transition between the dies and/or packages and the interposer that is not present in monolithic packages. This additional transition results in more impedance discontinuity with more opportunity for signal reflection. Such impedance discontinuities are particularly problematic as data rates increase.

[0017] In known PoINT architectures, the interposer serves to pass through signals between a processor die or package mounted on the interposer and other components (e.g., memory, peripheral component interconnect express (PCIe) connections, etc.) mounted on a circuit board with the signals also passing through a socket and the circuit board. Some example PoINT devices disclosed herein include at least some of the other components (e.g., memory, PCIe connections, etc.) with which the processor package is to communicate mounted directly onto the interposer with the processor package. In this manner, the processor package can communicate with these other components without the discontinuities defined by the transitions between the interposer and the socket, and between the socket and the circuit board. The processor package of some example PoINT devices disclosed herein also includes interconnects that bypass the interposer to directly connect with an underlying socket on a circuit board to communicate with other components via the circuit board. Bypassing the interposer in this manner eliminates the extra discontinuity that would otherwise result if an interposer was communicatively coupled between the processor package on the PoINT device and the socket on the circuit board. As a result, examples disclosed herein enable high speed connectivity by reducing discontinuities both by (1) positioning components directly on the interposer so that signals do not need to pass through a socket and circuit board, and (2) bypassing the interposer for signals that are to pass through the socket to the circuit board. Not only does this approach reduce discontinuities, it positions some of the other components (e.g., memory, PCIe connectors) closer to the processor package of the PoINT device (e.g., directly on the interposer) for improved performance.

[0018] FIG. **1** is a cross-sectional view of an example IC device assembly **100** constructed in accordance with teachings disclosed herein with an example package-on-interposer (PoINT) device **101** inserted into an example socket **116** on a circuit board **118** (e.g., a motherboard or other printed circuit board (PCB)). FIG. **2** is a cross-sectional view of the example IC device assembly **100** of FIG. **1** with the example PoINT device **101** spaced apart from the example socket **116**. In the illustrated example, the PoINT device **101** includes an integrated circuit (ICs) or semiconductor die or package **102** (e.g., a first semiconductor package), such as a processor or central processing unit (CPU package) that is coupled to a first side **104** of an interposer **106** via an array of contact pads or lands **108** (e.g., first contacts) on a mounting surface **110** (e.g., a bottom surface) of the semiconductor package **102**. In some examples, the semiconductor package **102** may include balls, pins, and/or pads, in addition to or instead of the contact pads **108**, to enable the electrical coupling of the semiconductor package **102** to the first side **104** of the interposer **106**. In some examples, the semiconductor package **102** includes a package substrate that provides mechanical support for a semiconductor die of the package **102** supported thereon. In some such examples, the package substrate provides an electrical interface between the semiconductor die mounted thereon and external components (e.g., the interposer **106** and the socket **116**). In some examples, the semiconductor package **102** does not include a package substrate. That is, in some examples, the semiconductor package **102** is simply a semiconductor die that is directly mounted on to the interposer **106**. In such examples, the interposer **106** serves the function of a package substrate. Thus, both interposers and package substrates are collectively referred to as substrates herein. Although similar in function, in some instances, the construction of the interposer **106** may differ from known package substrate technologies. For instance, unlike a package substrate, the interposer **106** of the illustrated example often has a more sophisticated structure, featuring a denser array of interconnects, vias, and through-silicon vias (TSVs) to facilitate high-bandwidth communication between the integrated components. Interposers enable dense integration of multiple components within a small footprint, allowing for more efficient use of space and improved system-level integration.

[0019] A package substrate generally includes a package core (that provides mechanical stability) and a build-up region on one or both sides of the core. The package core may be composed of purely organic or purely glass-based material. The build-up region(s) of a package substrate include layers of organic insulating material (e.g., laminated layers of prepreg) with intervening metal layers to define conductive traces and/or other metal features (e.g., ground planes) facilitating signal routing and power distribution.

[0020] An interposer can have a core layer (that provides mechanical stability) and insulating layers built on top and bottom of the core layer (e.g., in build-up regions) between layers of metal defining conductive traces and other metal features for signal routing and power distribution. Unlike the package core of a package substrate, which can be purely organic, or glass based and does not contain glass fiber weave, the interposer **106** of the illustrated example is constructed more like a printed circuit board. That is, in some examples, the core layer of an interposer can be composed of an organic material of epoxy resin material that is reinforced with woven glass fiber (e.g., flame retardant 4 (FR-4)). The core layer provides structural support and stability to the interposer. It serves as the main substrate. The FR-4 material offers electrical insulation, mechanical strength, and dimensional stability. The weave pattern of the glass fiber helps distribute mechanical stresses evenly across the interposer, reducing the risk of cracking or delamination. It also enhances thermal conductivity, aiding in heat dissipation. The woven glass fiber enhances the material's tensile strength and prevents warping. In some examples, the insulating material of the build-up regions(s) of the interposer **106** is an organic material (e.g., laminated layers of prepreg). Additionally or alternatively, in some examples, the build-up regions(s) of the interposer **106** may employ an epoxy resin with woven glass fiber (e.g., FR-4) similar to the core layer as the insulating material. The epoxy resin (in the core layer and/or the insulating layers) acts as a binder that holds the glass fiber weave together, forming a strong and durable composite material. Epoxy resin offers excellent adhesion to the glass fiber, ensuring a robust bond. The epoxy resin also provides electrical insulation properties and resistance to environmental factors such as moisture and chemicals.

[0021] The array of contact pads or lands **108** (e.g., ball grid array (BGA) pads) electrically couples to the interposer **106** via a ball grid array **109**. The ball grid array **109** is a mid-level interconnect between the semiconductor package **102** and the interposer **106**. In other examples, the interposer **106** includes an array of contact pads and the semiconductor package includes the interfacing ball grid array **109**. The array of contact pads **108** has a first shape **402**, as described below in connection with FIG. **4**. In this example, the semiconductor package **102** includes second contacts **112** on the mounting surface **110** of the semiconductor package **102**. The second contacts **112** are spaced apart from the first contacts (e.g., array of contact pads or lands **108**). The second contacts

112 enable the electrical coupling of the semiconductor package **102** to pins **140** (e.g., contacts) on the socket **116** that is mounted on top of the circuit board **118**. More particularly, in some examples, the second contacts **112** maybe a land grid array (LGA) pads that are urged against respective ones of the pins **140** when the PoINT device **101** is inserted into the socket **116** and engages (or at least comes into close contact with) a contact surface **114** (e.g., a seating plane) of the socket **116**. Thus, in this example, the second contacts **112** electrically coupled to the pins **140** define interconnects that bypass the interposer **106** to directly connect the semiconductor package **102** with the circuit board **118** through the underlying socket **116** to communicate with other components via the circuit board **118**. In some examples, the other components include an external component (e.g., PCIe connector **144**). The second contacts **112** have a second shape **404** that is different from the first shape **402** of the first contacts **108**, as described below in connection with FIG. **4**. As shown in FIG. **2**, when the PoINT device **101** is removed from the socket **116**, the pins **140** extend above the contact surface **114**.

[0022] As shown in the illustrated example, additional semiconductor dies or packages **120**, such as a memory (e.g., double data rate (DDR)) packages are also coupled to the first side **104** of the interposer **106** via arrays of contact pads or lands **122** on a mounting surface **124** (e.g., a bottom surface) of the additional (second) semiconductor packages **120**. The arrays of contact pads or lands **122** on the additional semiconductor packages **120** electrically couple to the interposer **106** via associated ball grid arrays **125** on the interposer. In other examples, the interposer **106** includes arrays of contact pads and the additional semiconductor package **120** includes interfacing ball grid arrays. In some examples, the additional semiconductor packages **120** may include balls, pins, and/or pads, in addition to or instead of the contact pads **122**, to enable the electrical coupling of the additional packages **120** to the first side **104** of the interposer **106**. In this example, the interposer **106** includes internal routing to electrically couple the first semiconductor package **102** to the additional semiconductor packages **120**. That is, individual balls in the first ball grid array **109** (connected to the first semiconductor package **102**) are electrically coupled through the interposer **106** to corresponding balls in the second ball grid arrays **125** (connected to the additional semiconductor packages **120**). The second ball grid arrays **125** are mid-level interconnects between the additional (second) semiconductor packages **120** and the interposer **106**. The first semiconductor package **102** is electrically coupled to the second semiconductor package **120** via the mid-level interconnect on the interposer **106**, and independent of the socket **116**. The first semiconductor package **102** (e.g., processor package) can communicate with the second semiconductor packages **120** (e.g., memory packages) without the discontinuities defined by the transitions between the interposer **106** and the socket **116**, and between the socket **116** and the circuit board **118**.

[0023] As noted above, in some examples, the interposer **106** is a substrate that includes glass, or organic materials (e.g., a core layer composed of an epoxy resin reinforced by woven fiberglass) and incorporates metal layers for electrical connections and signal routing. In some examples, the signal routing is to provide electrical interconnections between the separate semiconductor dies or semiconductor packages **102**, **120**. In some examples, the interposer **106** incorporates through-silicon vias (TSVs) for vertical connections between different layers. In some examples, the TSVs extend to and electrically connect with interconnects or contacts (e.g., pins) on and/or within the socket **116**. However, in other examples (such as the example shown in FIG. **1**), there is no direct electrical connection between the interposer **106** and the socket **116**. Thus, in some examples, TSVs may be omitted from the interposer **106**.

[0024] As shown in the illustrated example, the first side **104** of the interposer **106** is connected to the first contacts **108** of the first semiconductor package **102** and the contact pads **122** of the additional semiconductor packages **120**. The second side **126** of the interposer **106** is physically coupled to a contact surface **128** a first portion **129** of the socket **116**. However, in this example, the interposer **106** is devoid of electrical connections with the underlying socket **116**. In other examples, the second side **104** of the interposer **106** includes one or more contacts that electrical couple with the interposer **106** (e.g., with contacts similar to the pins **140** electrically couplable with the contacts **112** of the semiconductor package **102**. In some examples, the interposer **106** is to be between the first contacts **108** of the first semiconductor package **102** and the contact surface **128** of the socket **116**. Additionally, the interposer **106** is to be between the contact pads **122** of the second semiconductor packages **120** and the contact surface **128** of the socket **116**. In this example, the contact surface **128** is recessed relative to the contact surface **114** that is to engage the mounting surface **110** of the first semiconductor package **102**. In some examples, the socket **118** can be a zero-insertion-force (ZIF) socket, a land grid array (LGA) socket or any other type of socket depending on the specific requirements of the system. In some examples, the socket **118** includes a chamfer **132** to facilitate the alignment of the interposer **106** within the socket **118**. The interposer **106** may be connected to the socket **116** on the circuit board **118** utilizing solder, an adhesive, a compression (e.g., spring loaded) connectors, etc.

[0025] In the illustrated example of FIGS. **1** and **2**, the interposer **106** appears to be in two parts. However, in some examples, these parts are different portions of a single interposer connected at locations not shown in the cross-sectional view of FIGS. **1** and **2** but shown in the illustrated example of FIG. **3** discussed further below. More particularly, in some examples, the interposer **106** includes an opening or cutout **131** aligned with the second contacts **112** of the first semiconductor package **102** so that the interposer **106** is spaced apart from the second contacts **112** of the first semiconductor package **102**. In some examples, the second contacts **112** of the first semiconductor package **102** extend across (e.g., the contacts **112** face towards and/or face into) the opening or cutout **131** of the interposer **106**. Thus, as shown most clearly in FIG. **2**, the second contacts **112** are exposed through the opening **131** of the interposer **106**.

[0026] As shown in the illustrated example, a second portion **133** of the socket **116** is to extend into (e.g., through) the cutout **131** towards the first semiconductor package **102**. The second portion **133** of the socket **116** that extends through the cutout **131** in the interposer **106** includes the pins **140**. As shown in the illustrated example of FIG. **1**, when the PoINT device **101** is plugged or inserted into the socket **116**, the pins **140** electrically couple to the second contacts **112** of the first semiconductor package **102** exposed through the opening **131** of the interposer **106**. In this example, the interposer **106** is not between the second contacts **112** on the first semiconductor package **102** and the socket **116**. Whereas the second portion **133** of the socket **116** is constructed to fit through the cutout **131** of the interposer **130**, in some examples, the first portion **135** is to interface with (e.g., align with) the interposer **130**. More particularly, as noted above, the first portion **129** of the socket **116** includes and/or defines the contact surface **128**. In some examples, as shown most clearly in FIG. **3**, the first portion **129** surrounds the second portion **133**.

[0027] In some examples, the chamfer **132** on the socket **116** has a beveled or angled edge or surface **134**. The chamfer **132** aids in the alignment of the interposer **106** with the socket **116**. When the interposer **106** is inserted into the socket **116**, the chamfered edges **134** guide the interposer **106** into the correct position. In the illustrated example, although the interposer **106** is inserted into the socket **116**, the interposer **106** is not electrically coupled with the socket **116**. However, the interposer **106** still needs to be aligned with the socket **116** so that the second contacts **112** on the first semiconductor package **102** (attached to the interposer **106**) are aligned with corresponding pins **140** of the socket **106**. Inasmuch as the pins **140** extend through the cutout **131** of the interposer **106**, the first semiconductor package **102** is able to electrically couple with the socket **116** independent of the interposer **106**.

[0028] The socket **116** is coupled to a first face **136** of the circuit board **118** by coupling components **138** (e.g., electrical contacts or interconnects). The coupling components **138** may electrically and mechanically couple the socket **116** to the circuit board **118**, and may include solder balls (as shown in FIG. **1**), and/or any other suitable electrical and/or mechanical coupling structure. In the illustrated example, the electrical connections between the socket **116** and the circuit board **118** are aligned with the second portion **133** of the socket **116** and spaced apart from the first portion **129** of the socket **116**. The coupling components **138** electrically couple the first semiconductor package **102** via the pins **140** in the second portion **133** of the socket **116** to interconnects **142** (e.g., electrical traces or other routing) in the circuit board **118**. In some examples, the socket **116** includes one or more standoffs **143** which may assist in supporting the socket **116** with the circuit board **118**, ensuring proper electrical connections and preventing misalignment issues during installation. In some examples, the standoffs **143** are an integral component of the socket **116**. In other examples, the standoffs **143** are distinct and separate from the socket **116**. In some examples, the standoffs **143** may be omitted.

[0029] The IC device assembly **100** illustrated in FIG. **1** includes a PCIe connector **144** coupled to the first face **136** of the circuit board **118**. The connector **144** is coupled to the coupling components **138** via the interconnects **142**. The connector **144** couples the circuit board **118** with a cable that connects to a peripheral component interconnect express (PCIe) modules or devices.

[0030] To enable the electrical coupling of the various components of the example IC device assembly **100** of FIG. **1**, consideration needs to be given to the thicknesses of the different components. For instance, the mid-level interconnects corresponding to the ball grid arrays **109**, **125** between the semiconductor packages **102**, **120** and the interposer **106** have a first thickness **146**. Using known interconnect technology, the first thickness **146** is approximately 0.257 millimeter (mm). However, in other examples, the first thickness **146** can be more or less than this distance. Further, the interposer **106** has a second thickness **148** that is largely driven by the number of metal layers needed to implement all electrical routing between components to be interconnected. Some interposers in known PoINT devices include as many as 20 metal routing layers which results in a thickness of approximately 1.6 mm. However, in such known PoINT devices, the interposer includes routing for PCIe signals. Unlike such known PoINT devices, the interposer **106** of the example PoINT device **101** does not need any metal routing for PCIe signals because such signals bypass the interposer **106** and pass directly from the first semiconductor package **102** to the socket **116**. The PCIe signals are routed through the pins **140** in the socket **116** without passing through the interposer **106**, thereby reducing the number of routing layers needed in the interposer **106**. Accordingly, in some examples, the interposer **106** can have significantly fewer metal layers and, thus, can have a thickness **148** significantly less than 1.6 mm. Specifically, in some examples, the thickness **148** can be less than or equal to 1.4 mm, less than or equal to 1.2 mm, less than or equal to 1.0 mm, etc. By routing PCIe signals directly through the pins **140** without going through the interposer **106**, the PCIe signals have one less mid-level interconnect interface to pass through. This reduces the number of impedance discontinuities for communications with external components (e.g., PCIe connected devices) which results in less signal reflections and improves signal integrity (SI). Maintaining signal integrity in communications ensures reliable data transmission at high speed.

[0031] The combined distance of the first thickness **146** and the second thickness **148** corresponds to the approximate distance the contact surface **128** of the socket **106** needs to be spaced apart from the bottom surface **110** of the semiconductor package **102**. In some examples, the distance between the bottom surface **110** of the semiconductor package **102** and the contact surface **128** of the socket **116** is slightly greater than the combined distance of the first and second thicknesses **146**, **148** to ensure the mid-level interconnects (e.g., the ball grid arrays **109**, **125**) do not get crushed during insertion of the PoINT device **101** into the socket **106**. More particularly, as shown in FIG. **1**, the distance between the semiconductor package **102** and the contact surface **128** of the socket **116** corresponds to the first thickness **146**, the second thickness **148**, and a third thickness **150**. In some examples, the third thickness **150** is relatively small (e.g., 0.3 mm or less, 0.2 mm or less, 0.15 mm or less, 0.14 mm or less, 0.1 mm or less, etc.). In some examples, the extra distance corresponding to the third thickness **150** is filled by a compressible material **130** positioned between the interposer **106** and the contact surface **128** of the socket **116**. In some examples, the compressible material

130 has a thickness larger than the third thickness **150** when the compressible material **130** is uncompressed (as shown in FIG. **2**) but that compresses down to the third thickness **150** when the PoINT device **101** is inserted into the socket **116** (as shown in FIG. **1**).

[0032] In some examples, the compressible material **130** is attached (e.g., adhered) to the second side **126** of the interposer **106**. In some examples, the compressible material **130** is attached (e.g., adhered) to the contact surface **128** of the socket **116**. In some examples, the compressible material **130** can be a silicon rubber with an example hardness scale of **40**A durometer. However, any other suitable material may also be used. In some examples, the first portion **129** of the socket **116** (e.g., the portion associated with the contact surface **128**) can have a first thickness **152** and the second portion **133** of the socket **116** can have a second thickness **154**. As shown, the first thickness **152** is less than the second thickness **154**. In some examples, as shown in the illustrated example, the first and second portions **129**, **133** share a common bottom surface **156** (e.g., the underside, board facing surface) of the socket **116** (e.g., the first and second portions **129**, **133** are flush with one another along an underside of the socket **116**). In other examples, the first and second portions **129**, **133** can be offset relative to one another along the bottom surface **156** of the socket **116**. In the illustrated example, the first portion **129** is positioned between the semiconductor dies and/or packages **102**, **120** and the standoffs **143** on the circuit board **118** when the semiconductor die **102**, **120** is coupled to the socket. In this example, the first thickness **152** corresponds to the distance between the contact surface **128** on the socket **116** and the bottom surface **156** of the socket **116**. In some examples, the first thickness **152** is less than or equal to 1 mm (e.g., less than or equal to 0.75 mm, less than or equal to 0.6 mm, less than or equal to 0.5 mm, less than or equal to 0.4 mm, etc.).

[0033] In the illustrated example, the second portion **133** of the socket **116** is positioned between the semiconductor die or package **102** and the circuit board **118** when the semiconductor package **102** is coupled to the socket **116**. The second portion **133** of the socket **116** includes the pins **140**. The pins **140** extend through the second thickness **154** of the second portion of the socket **116**. The second portion **133** of the socket **116** is not between the interposer **106** and the circuit board **118** when the interposer **106** is inserted into the socket **116**. This is because the second portion **133** of the socket **116** extends through the opening or cutout **131** of the interposer **106**. Hence, the first thickness **152** of the first portion **129** of the socket **116** is less than the second thickness **154** of the second portion **133** of the socket **116**. More particularly, in some examples, the difference between the first thickness **152** and the second thickness **154** corresponds to the combined thickness of the thickness **146** of the balls in the ball grid arrays **109**, **125**, the thickness **148** of the interposer **106**, and the thickness **150** of the compressible material **130** (when compressed) as described above. In this example, the second thickness **154** corresponds to the distance between the upper contact surface **114** on the socket **116** and the bottom surface **156** of the socket **116**. In some examples, the second thickness **154** is less than or equal to 3 mm (e.g., less than or equal to 2.5 mm, less than or equal to 2 mm, less than or equal to 1. mm, etc.). In some examples, the socket **116** has an overall height **158** that corresponds to the combined thickness associated with the second thickness **154** of the second portion **133** and the thickness or height of the standoffs **143** that define the gap between the bottom surface **156** and the circuit board **118**.

[0034] FIG. **3** illustrates an exploded view of the example IC device assembly **100** of FIG. **1**. The first semiconductor package **102** is coupled to the interposer **106** that includes an opening or a cutout **131**. The opening **131** extends from the first side **104** of the interposer **106** to the second side **126** of the interposer **106**. The first side **104** is opposite the second side **126** of the interposer **106**. The first contacts **108** on the bottom surface of the semiconductor package **102** couple to the ball grid array **109** on the interposer **106**. In this example, the ball grid array **109** is adjacent to the cutout **131**. More particularly, in this example, the ball grid array **109** surrounds the cutout **131**. In other examples, the ball grid array **109** may adjacent some of the side of the cutout **131** but not all of them. The interposer **106** is dimensioned to be inserted into the socket **116** (e.g., adjacent the contact surface **128** associated with the first portion **129** of the socket **116**). The second side **126** of the interposer **106** faces towards the socket **116**. The socket **116** is mounted on top of the circuit board **118**. As shown in the illustrated example, the semiconductor package **102** extends across (e.g., covers) the opening or cutout **131** when the package **102** is coupled to the interposer **106** via the ball grid array **108**. The second contacts **112** of the semiconductor package **102** electrically couple to the circuit board **118** via the pins **140** in the second portion **133** of the socket **116** that is dimensioned to extend through the opening or cutout **131** to reach the underside of the semiconductor package **102** where the second contacts **112** are located (as shown in FIGS. **1** and **2**). In this arrangement, the interconnects made by the electrical coupling of the second contacts **112** with the pins **140** bypass the interposer **106**. The interposer **106** is spaced apart from the pins **140**. More particularly, as shown in the illustrated example, the interposer **106** is dimensioned to extend around the second portion **133** of the socket **116** where the pins **140** are located. In some examples, the interposer **106** may not extend completely around the second portion **133**. For instance, in some examples, the interposer **133** may be generally "C" shaped with the opening or cutout **131** extending to an edge or perimeter of the interposer **106**. In other examples, the interposer **106** may include more than one discrete section that are spaced apart from one another (e.g., positioned on opposite sides of the second portion **133** of the socket **116**.

[0035] In the illustrated example of FIG. **3**, one or more second semiconductor packages **120** (e.g., memory package) are coupled to the interposer **106** on the first side or surface **104** of the interposer **106** (e.g., via the ball grid arrays **125** shown in FIGS. **1** and **2**). In some examples, the second semiconductor package **120** are electrically coupled to the processor package **102** through the interposer **106**, and independent of the socket **116**. The connection between the semiconductor packages **102**, **120** through the interposer **106** via the mid-level interconnect has a shorter interconnect compared to a conventional architecture which has an interconnect through the interposer **106** and the socket **116**, and between the socket **116** and the circuit board **118**. This shorter interconnect length improves performance of artificial intelligence (AI) and/or high performance computing (HPC) applications.

[0036] In some examples, as shown, the second portion **133** of the socket **116** includes an opening or recess **202** to provide space for components (e.g., a power deliver capacitor) attached to the underside of the semiconductor package **102**.

[0037] FIG. **4** is a bottom view of the example semiconductor package **102** of the example PoINT device **101** of FIGS. **1-3**. In this example, the semiconductor package **102** includes the first contacts **108** (e.g., contact pads) and the second contacts **112** on the mounting surface **110** of the semiconductor package **102**. As shown in the illustrated example, ones of the first contact pads **108** have a first shape **402** that is different from a second shape **404** of the second contact pads **112**. In some examples, the different shapes **402**, **404** are based on the different modes of electrical coupling with the adjacent components (e.g., the interposer **106** and the socket **116**. More particularly, in this example, the first contact pads **108** are ball grid array pads having a first shape **402** that is circular and the second contacts **112** having a second shape **404** that is rectangular. In other examples, the first contact pads **108** and/or the second contact pads **112** can be any other suitable shape. In some examples, the first contact pads **108** and the second contacts pads **112** have the same shape. In some such examples, the pads **108**, **112** may nevertheless be different in size.

[0038] FIG. **5** is a cross-sectional view of another example IC device assembly **500** in accordance with teachings disclosed herein. The example IC device assembly **500** includes an example PoINT device **501** that is similar to the PoINT device **101** of FIGS. **1-3**. Accordingly, the same reference numbers will be used for the same or similar features and the descriptions of such features provided above in connection with FIGS. **1-4** apply equally to the corresponding features shown in the illustrated example of FIG. **5**. The example PoINT device **501** of FIG. **5** differs from the example PoINT device **101** of FIGS. **1-3** in that the interposer **106** is structured to provide space to support an example peripheral component connector **502** mounted thereon in addition to the second semiconductor packages **120**. In this example, the peripheral component connector **502** is coupled to the first side **104** of the interposer **106** via an array of contact pads or lands **504** on the bottom surface of the peripheral component connector **502**. The array of contact pads or lands **504** on the peripheral component connector **502** electrically couple to the interposer **106** via an associated ball grid array **506** on the interposer **106**. In some examples, the peripheral component connector **502** is electrically coupled to one or more of the semiconductor packages **102**, **120** via the interposer **106** and independent of the socket **116**. In this example, the peripheral component connector **502** is a peripheral component interconnect express (PCIe) connector. However, in other examples, the peripheral component connector **502** can be any other suitable component connector.

[0039] The example IC device assembly **500** of FIG. **5** differs from the example IC device assembly **100** of FIGS. **1-3** in that the standoffs **146** adjacent the circuit board **118** are omitted. Rather, in the illustrated example of FIG. **5**, the first face **136** of the circuit board **118** includes one or more dummy bumps **508**. As used herein, dummy bumps are bumps that are not used to carry electrical signals. In this example, the dummy bumps **508** provide structural support to the underside of the socket **116** (hence, the reason the standoffs **143** shown in FIGS. **1** and **2** are omitted). In some examples, the dummy bumps **508** are similarly sized and shaped to the coupling components **138** (e.g., electrical contacts or interconnects) used to electrically and mechanically couple the socket **116** to the circuit board **118** and enable the electrical coupling of the semiconductor package **102** to the circuit board **118** through the socket **116** (and independent of the interposer **106**). In some examples, the dummy bumps **508** are disposed on the first face **136** of the circuit board **118**.

[0040] FIG. **6** is a cross-sectional view of another example IC device assembly **600** in accordance with teachings disclosed herein. The example IC device assembly **600** includes an example PoINT device **601** that is similar to the PoINT device **501** of FIG. **5** (and, by extension, similar to the PoINT device **101** of FIGS. **1-3**). Accordingly, the same reference numbers will be used for the same or similar features and the descriptions of such features provided above in connection with FIGS. **1-5** apply equally to the corresponding features shown in the illustrated example of FIG. **6**. The example PoINT device **601** of FIG. **5** differs from the example PoINT devices **101**, **501** of FIGS. **1-5** in that the interposer **106** is structured to provide space to support an electronic integrated circuit (EIC) **606** that is electrically coupled with a photonic integrated circuit (PIC) **604**. More particularly, in this example, the PIC **604** is stacked on top of the EIC **606**. In some examples, an optical fiber cable **602** is coupled to the photonic integrated circuit (PIC) **604** to enable optical communication with an external component. In this example, the PIC **604** includes contact pads **608**. In some examples, the PIC **604** may include balls, pins, and/or pads, in addition to or instead of the contact pads **608**, to enable the electrical coupling of the PIC **604** to the EIC **606**. The array of contact pads **608** (e.g., ball grid array pads) electrically couples to the EIC **606** via a ball grid array **610**. The EIC **606** includes contact pads **612**. In some examples, the EIC **606** may include balls, pins, and/or pads, in addition to or instead of the contact pads **612**, to enable the electrical coupling of the EIC **606** to the interposer **106**. The array of contact pads **612** (e.g., ball grid array pads) electrically couples to the first side **104** of the interposer **106** via a ball grid array **614**.

[0041] FIG. **7** is a flowchart representative of an example method to manufacture any one of the example IC device assemblies **100**, **500**, **600** of FIGS. **1-6**. The operations **700** begin at block **702**, at which the interposer **106** (FIG. **1**) with the cutout **131** (FIG. **1**) is created. At block **704**, semiconductor die(s) **102**, **120** (FIG. **1**) are mounted onto the interposer **106**. At block **706**, the socket **116** (FIG. **1**) is created to receive the interposer **106** and extend through the cutout **131** of the interposer **106**. In some examples, block **706** can be implemented independent of (e.g., parallel with and/or before) blocks **702** and **704**. At block **708**, the socket **116** is mounted on the circuit board **118**. In some examples, block **708** can also be implemented independent of (e.g., parallel with and/or before) blocks **702** and **704**. At block **710**, the interposer **106** is aligned and placed in the socket **116**. At block **712**, a load is applied to the IC device assemblies of FIGS. **1-6** to establish electrical coupling of the semiconductor die(s) **102**, **120** to the socket **116**. In some examples, a load is applied because the interposer **106** is compression-mounted interconnect devices that are positioned between the socket **116** or circuit board **118** to allow an electrical signal to pass through.

[0042] FIG. **8** is a top view of a wafer **800** and dies **802** that may be included in an IC package (e.g., as discussed below with reference to FIG. **10**) in accordance with any of the examples disclosed herein (e.g., the dies **802** may correspond to any one of the example semiconductor packages **102**, **120** of FIGS. **1-8**, the connector **502** of FIG. **5**, the PIC **604** of FIG. **6**, and/or the EIC **606** of FIG. **6**). The wafer **800** includes semiconductor material and one or more dies **802** having circuitry. Each of the dies **802** may be a repeating unit of a semiconductor product. After the fabrication of the semiconductor product is complete, the wafer **800** may undergo a singulation process in which the dies **802** are separated from one another to provide discrete "chips." The die **802** includes one or more transistors (e.g., some of the transistors **940** of FIG. **9**, discussed below), supporting circuitry to route electrical signals to the transistors, passive components (e.g., traces, resistors, capacitors, inductors, and/or other circuitry), and/or any other components. In some examples, the die **802** may include and/or implement a memory device (e.g., a random access memory (RAM) device, such as a static RAM (SRAM) device, a magnetic RAM (MRAM) device, a resistive RAM (RRAM) device, a conductive-bridging RAM (CBRAM) device, etc.), a logic device (e.g., an AND, OR, NAND, or NOR gate), or any other suitable circuitry or electronics. Multiple ones of these devices may be combined on a single die **802**. For example, a memory array of multiple memory circuits may be formed on a same die **802** as programmable circuitry (e.g., the processor circuitry **1202** of FIG. **12**) and/or other logic circuitry. Such memory may store information for use by the programmable circuitry.

[0043] FIG. **9** is a cross-sectional side view of an IC device **900** that may be included in an IC package (e.g., as discussed below with reference to FIG. **10**), in accordance with any of the examples disclosed herein (e.g., the device **900** may correspond to any one of the example semiconductor packages **102**, **120** of FIGS. **1-8**, the connector **502** of FIG. **5**, the PIC **604** of FIG. **6**, and/or the EIC **606** of FIG. **6**). One or more of the IC devices **900** may be included in one or more dies **802** (FIG. **8**). The IC device **900** may be formed on a die substrate **902** (e.g., the wafer **800** of FIG. **8**) and may be included in a die (e.g., the die **802** of FIG. **8**). The die substrate **902** may be a semiconductor substrate including semiconductor materials including, for example, n-type or p-type materials systems (or a combination of both). The die substrate **902** may include, for example, a crystalline substrate formed using a bulk silicon or a silicon-on-insulator (SOI) substructure. In some examples, the die substrate **902** may be formed using alternative materials, which may or may not be combined with silicon, that include but are not limited to germanium, indium antimonide, lead telluride, indium arsenide, indium phosphide, gallium arsenide, or gallium antimonide. Further materials classified as group II-VI, III-V, or IV may also be used to form the die substrate **902**. Although a few examples of materials from which the die substrate **902** may be formed are described here, any material that may serve as a foundation for an IC device **900** may be used. The die substrate **902** may be part of a singulated die (e.g., the dies **802** of FIG. **8**) or a wafer (e.g., the wafer **800** of FIG. **8**).

[0044] The IC device **900** may include one or more device layers **904** disposed on and/or above the die substrate **902**. The device layer **904** may include features of one or more transistors **940** (e.g., metal oxide semiconductor field-effect transistors (MOSFETs)) formed on the die substrate **902**. The device layer **904** may include, for example, one or more source and/or drain (S/D) regions **920**, a gate **922** to control current flow between the S/D regions **920**, and one or more S/D contacts **924** to route electrical signals to/from the S/D regions **920**. The transistors **940** may include additional features not depicted for the sake of clarity, such as device isolation regions, gate contacts, and the like. The transistors **940** are not limited to the type and configuration depicted in FIG. **9** and may include a wide variety of other types and/or configurations such as, for example, planar transistors, non-planar transistors, or a combination of both. Non-planar transistors may include FinFET transistors, such as double-gate transistors or tri-gate transistors, and wrap-around or all-around gate transistors, such as nanoribbon and nanowire transistors.

[0045] Each transistor **940** may include a gate **922** including a gate dielectric and a gate electrode. The gate dielectric may include one layer or a stack of layers. The one or more layers may include silicon oxide, silicon dioxide, silicon carbide, and/or a high-k dielectric material. The high-k dielectric material may include elements such as hafnium, silicon, oxygen, titanium, tantalum, lanthanum, aluminum, zirconium, barium, strontium, yttrium, lead, scandium, niobium, and/or zinc. Examples of high-k materials that may be used in the gate dielectric include, but are not limited to, hafnium oxide, hafnium silicon oxide, lanthanum oxide, lanthanum aluminum oxide, zirconium oxide, zirconium silicon oxide, tantalum oxide, titanium oxide, barium strontium titanium oxide, barium titanium oxide, strontium titanium oxide, yttrium oxide, aluminum oxide, lead scandium tantalum oxide, and/or lead zinc niobate. In some examples, an annealing process may be carried out on the gate dielectric to improve its quality when a high-k material is used.

[0046] The gate electrode may be formed on the gate dielectric and may include at least one p-type work function metal or n-type work function metal, depending on whether the transistor **940** is to be a p-type metal oxide semiconductor (PMOS) or an n-type metal oxide semiconductor (NMOS) transistor. In some implementations, the gate electrode may include a stack of two or more metal layers, where one or more metal layers are work function metal layers and at least one metal layer is a fill metal layer. Further metal layers may be included, such as a barrier layer. For a PMOS transistor, metals that may be used for the gate electrode include, but are not limited to, ruthenium, palladium, platinum, cobalt, nickel, conductive metal oxides (e.g., ruthenium oxide), and/or any of the metals discussed below with reference to an NMOS transistor (e.g., for work function tuning). For an NMOS transistor, metals that may be used for the gate electrode include, but are not limited to, hafnium, zirconium, titanium, tantalum, aluminum, alloys of these metals, carbides of these metals (e.g., hafnium carbide, zirconium carbide, titanium carbide, tantalum carbide, and/or aluminum carbide), and/or any of the metals discussed above with reference to a PMOS transistor (e.g., for work function tuning).

[0047] In some examples, when viewed as a cross-section of the transistor **940** along the source-channel-drain direction, the gate electrode may include a U-shaped structure that includes a bottom portion substantially parallel to the surface of the die substrate **902** and two sidewall portions that are substantially perpendicular to the top surface of the die substrate **902**. In other examples, at least one of the metal layers that form the gate electrode may be a planar layer that is substantially parallel to the top surface of the die substrate **902** and does not include sidewall portions substantially perpendicular to the top surface of the die substrate **902**. In other examples, the gate electrode may include a combination of U-shaped structures and/or planar, non-U-shaped structures. For example, the gate electrode may include one or more U-shaped metal layers formed atop one or more planar, non-U-shaped layers.

[0048] In some examples, a pair of sidewall spacers may be formed on opposing sides of the gate stack to bracket the gate stack. The sidewall spacers may be formed from materials such as silicon nitride, silicon oxide, silicon carbide, silicon nitride doped with carbon, and/or silicon oxynitride. Processes for forming sidewall spacers are well known in the art and generally include deposition and etching process operations. In some examples, a plurality of spacer pairs may be used; for instance, two pairs, three pairs, or four pairs of sidewall spacers may be formed on opposing sides of the gate stack.

[0049] The S/D regions **920** may be formed within the die substrate **902** adjacent to the gate **922** of corresponding transistor(s) **940**. The S/D regions **920** may be formed using an implantation/diffusion process or an etching/deposition process, for example. In the former process, dopants such as boron, aluminum, antimony, phosphorous, or arsenic may be ion-implanted into the die substrate **902** to form the S/D regions **920**. An annealing process that activates the dopants and causes them to diffuse farther into the die substrate **902** may follow the ion-implantation process. In the latter process, the die substrate **902** may first be etched to form recesses at the locations of the S/D regions **920**. An epitaxial deposition process may then be carried out to fill the recesses with material that is used to fabricate the S/D regions **920**. In some implementations, the S/D regions **920** may be fabricated using a silicon alloy such as silicon germanium or silicon carbide. In some examples, the epitaxially deposited silicon alloy may be doped in situ with dopants such as boron, arsenic, or phosphorous. In some examples, the S/D regions **920** may be formed using one or more alternate semiconductor materials such as germanium or a group III-V material or alloy. In further examples, one or more layers of metal and/or metal alloys may be used to form the S/D regions **920**.

[0050] Electrical signals, such as power and/or input/output (I/O) signals, may be routed to and/or from the devices (e.g., transistors **940**) of the device layer **904** through one or more interconnect layers disposed on the device layer **904** (illustrated in FIG. **9** as interconnect layers **906-910**). For example, electrically conductive features of the device layer **904** (e.g., the gate **922** and the S/D contacts **924**) may be electrically coupled with the interconnect structures **928** of the interconnect layers **906-910**. The one or more interconnect layers **906-910** may form a metallization stack (also referred to as an “ILD stack”) **919** of the IC device **900**.

[0051] The interconnect structures **928** may be arranged within the interconnect layers **906-910** to route electrical signals according to a wide variety of designs (in particular, the arrangement is not limited to the particular configuration of interconnect structures **928** depicted in FIG. **9**). Although a particular number of interconnect layers **906-910** is depicted in FIG. **9**, examples of the present disclosure include IC devices having more or fewer interconnect layers than depicted.

[0052] In some examples, the interconnect structures **928** may include lines **928***a* and/or vias **928***b* filled with an electrically conductive material such as a metal. The lines **928***a* may be arranged to route electrical signals in a direction of a plane that is substantially parallel with a surface of the die substrate **902** upon which the device layer **904** is formed. For example, the lines **928***a* may route electrical signals in a direction in and/or out of the page from the perspective of FIG. **9**. The vias **928***b* may be arranged to route electrical signals in a direction of a plane that is substantially perpendicular to the surface of the die substrate **902** upon which the device layer **904** is formed. In some examples, the vias **928***b* may electrically couple lines **928***a* of different interconnect layers **906-910** together.

[0053] The interconnect layers **906-910** may include a dielectric material **926** disposed between the interconnect structures **928**, as shown in FIG. **9**. In some examples, the dielectric material **926** disposed between the interconnect structures **928** in different ones of the interconnect layers **906-910** may have different compositions; in other examples, the composition of the dielectric material **926** between different interconnect layers **906-910** may be the same.

[0054] A first interconnect layer **906** (referred to as Metal 1 or “M1”) may be formed directly on the device layer **904**. In some examples, the first interconnect layer **906** may include lines **928***a* and/or vias **928***b*, as shown. The lines **928***a* of the first interconnect layer **906** may be coupled with contacts (e.g., the S/D contacts **924**) of the device layer **904**.

[0055] A second interconnect layer **908** (referred to as Metal 2 or “M2”) may be formed directly on the first interconnect layer **906**. In some examples, the second interconnect layer **908** may include vias **928***b* to couple the lines **928***a* of the second interconnect layer **908** with the lines **928***a* of the first interconnect layer **906**. Although the lines **928***a* and the vias **928***b* are structurally delineated with a line within each interconnect layer (e.g., within the second interconnect layer **908**) for the sake of clarity, the lines **928***a* and the vias **928***b* may be structurally and/or materially contiguous (e.g., simultaneously filled during a dual-damascene process) in some examples.

[0056] A third interconnect layer **910** (referred to as Metal 3 or “M3”) (and additional interconnect layers, as desired) may be formed in succession on the second interconnect layer **908** according to similar techniques and/or configurations described in connection with the second interconnect layer **908** or the first interconnect layer **906**. In some examples, the interconnect layers that are “higher up” in the metallization stack **919** in the IC device **900** (i.e., further away from the device layer

904) may be thicker.

[0057] The IC device **900** may include a solder resist material **934** (e.g., polyimide or similar material) and one or more conductive contacts **936** formed on the interconnect layers **906-910**. In FIG. **9**, the conductive contacts **936** are illustrated as taking the form of bond pads. The conductive contacts **936** may be electrically coupled with the interconnect structures **928** and configured to route the electrical signals of the transistor(s) **940** to other external devices. For example, solder bonds may be formed on the one or more conductive contacts **936** to mechanically and/or electrically couple a chip including the IC device **900** with another component (e.g., a circuit board). The IC device **900** may include additional or alternate structures to route the electrical signals from the interconnect layers **906-910**; for example, the conductive contacts **936** may include other analogous features (e.g., posts) that route the electrical signals to external components.

[0058] FIG. **10** is a cross-sectional view of an example IC package **1000** constructed in accordance with teachings disclosed herein. In some examples, the IC package **1000** may correspond to any one of the example semiconductor packages **102**, **120** of FIGS. **1-8**, the connector **502** of FIG. **5**, the PIC **604** of FIG. **6**, and/or the EIC **606** of FIG. **6**. The package substrate **1002** may include a dielectric material and may have conductive pathways extending through the dielectric material between upper and lower faces **1022**, **1024**, and/or between different locations on the upper face **1022**, and/or between different locations on the lower face **1024**. These conductive pathways may take the form of any of the interconnects **928** discussed above with reference to FIG. **9**.

[0059] The IC package **1000** may include a die **1006** coupled to the package substrate **1002** via conductive contacts **1004** of the die **1006**, first-level interconnects **1008**, and conductive contacts **1010** of the package substrate **1002**. The conductive contacts **1010** may be coupled to conductive pathways **1012** through the package substrate **1002**, allowing circuitry within the die **1006** to electrically couple to various ones of the conductive contacts **1014**. The first-level interconnects **1008** illustrated in FIG. **10** are solder bumps, but any suitable first-level interconnects **1008** may be used. As used herein, a “conductive contact” refers to a portion of conductive material (e.g., metal) serving as an electrical interface between different components. Conductive contacts may be recessed in, flush with, or extending away from a surface of a component, and may take any suitable form (e.g., a conductive pad or socket).

[0060] In some examples, an underfill material **1016** may be disposed between the die **1006** and the package substrate **1002** around the first-level interconnects **1008**, and/or a mold compound **1018** may be disposed around the die **1006** and in contact with the package substrate **1002**. In some examples, the underfill material **1016** may be the same as the mold compound **1018**. Example materials that may be used for the underfill material **1016** and the mold compound **1018** are epoxy mold materials, as suitable. Second-level interconnects **1020** may be coupled to the conductive contacts **1014**. The second-level interconnects **1020** illustrated in FIG. **10** are solder balls (e.g., for a ball grid array arrangement), but any suitable second-level interconnects **1020** may be used (e.g., pins in a pin grid array arrangement or lands in a land grid array arrangement). The second-level interconnects **1020** may be used to couple the IC package **1000** to another component, such as a circuit board (e.g., a motherboard), an interposer, or another IC package, as known in the art and as discussed below with reference to FIG. **11**.

[0061] In FIG. **10**, the IC package **1000** is a flip chip package, and includes a package substrate **1002**. The number and location of the package substrate **1002** of the IC package **1000** is simply illustrative. The die **1006** may take the form of any of the examples of the die **1202** discussed herein (e.g., may include any of the examples of the IC device **900**).

[0062] Although the IC package **1000** illustrated in FIG. **10** is a flip chip package, other package architectures may be used. For example, the IC package **1000** may be a ball grid array (BGA) package, such as an embedded wafer-level ball grid array (eWLB) package. In another example, the IC package **1000** may be a wafer-level chip scale package (WLCSP) or a panel fanout (FO) package. Although a single die **1006** is illustrated in the IC package **1000** of FIG. **10**, an IC package **1000** may include multiple dies **1006** (e.g., with one or more of the multiple dies **1006** coupled to the package substrate **1002**). An IC package **1000** may include additional passive components, such as surface-mount resistors, capacitors, and/or inductors disposed on the first face **1022** or the second face **1024** of the package substrate **1002**. More generally, an IC package **1000** may include any other active and/or passive components known in the art.

[0063] FIG. **11** is a cross-sectional side view of an IC device assembly **1100** disclosed herein. In some examples, the IC device assembly **1100** includes and/or corresponds to any one of the example IC device assemblies **100**, **500**, **600** of FIGS. **1-6**, and/or any one of the associated examples PoINT devices **101**, **501**, **601** of FIGS. **1-6**. The IC device assembly **1100** includes a number of components disposed on a circuit board **1102** (which may be, for example, a motherboard). The IC device assembly **1100** includes components disposed on a first face **1140** of the circuit board **1102** and an opposing second face **1142** of the circuit board **1102**; generally, components may be disposed on one or both faces **1140** and **1142**.

[0064] In some examples, the circuit board **1102** may be a printed circuit board (PCB) including multiple metal layers separated from one another by layers of dielectric material and interconnected by electrically conductive vias. Any one or more of the metal layers may be formed in a desired circuit pattern to route electrical signals (optionally in conjunction with other metal layers) between the components coupled to the circuit board **1102**. In other examples, the circuit board **1102** may be a non-PCB substrate.

[0065] The IC device assembly **1100** illustrated in FIG. **11** includes a package-on-interposer structure **1136** coupled to the first face **1140** of the circuit board **1102** by coupling components **1116**. The coupling components **1116** may electrically and mechanically couple the package-on-interposer structure **1136** to the circuit board **1102**, and may include solder balls (as shown in FIG. **11**), male and female portions of a socket, an adhesive, an underfill material, and/or any other suitable electrical and/or mechanical coupling structure.

[0066] The package-on-interposer structure **1136** may include an IC package **1120** coupled to an interposer **1104** by coupling components **1118**. The coupling components **1118** may take any suitable form for the application, such as the forms discussed above with reference to the coupling components **1116**. Although a single IC package **1120** is shown in FIG. **11**, multiple IC packages may be coupled to the interposer **1104**; indeed, additional interposers may be coupled to the interposer **1104**. The interposer **1104** may provide an intervening substrate used to bridge the circuit board **1102** and the IC package **1120**. The IC package **1120** may be or include, for example, a die (the die **802** of FIG. **8**), an IC device (e.g., the IC device **900** of FIG. **9**), or any other suitable component. Generally, the interposer **1104** may spread a connection to a wider pitch or reroute a connection to a different connection. For example, the interposer **1104** may couple the IC package **1120** (e.g., a die) to a set of BGA conductive contacts of the coupling components **1116** for coupling to the circuit board **1102**. In the example illustrated in FIG. **11**, the IC package **1120** and the circuit board **1102** are attached to opposing sides of the interposer **1104**; in other examples, the IC package **1120** and the circuit board **1102** may be attached to a same side of the interposer **1104**. In some examples, three or more components may be interconnected by way of the interposer **1104**.

[0067] In some examples, the interposer **1104** may be formed as a PCB, including multiple metal layers separated from one another by layers of dielectric material and interconnected by electrically conductive vias. In some examples, the interposer **1104** may be formed of an epoxy resin, a fiberglass-reinforced epoxy resin, an epoxy resin with inorganic fillers, a ceramic material, or a polymer material such as polyimide. In some examples, the interposer **1104** may be formed of alternate rigid or flexible materials that may include the same materials described above for use in a semiconductor substrate, such as silicon, germanium, and other group III-V and group IV materials. The interposer **1104** may include metal interconnects **1108** and vias **1110**, including but not limited to through-silicon vias (TSVs) **1106**. The interposer **1104** may further include embedded devices

1114, including both passive and active devices. Such devices may include, but are not limited to, capacitors, decoupling capacitors, resistors, inductors, fuses, diodes, transformers, sensors, electrostatic discharge (ESD) devices, and memory devices. More complex devices such as radio frequency devices, power amplifiers, power management devices, antennas, arrays, sensors, and microelectromechanical systems (MEMS) devices may also be formed on the interposer **1104**. The package-on-interposer structure **1136** may take the form of any of the package-on-interposer structures known in the art.

[0068] The IC device assembly **1100** may include an IC package **1124** coupled to the first face **1140** of the circuit board **1102** by coupling components **1122**. The coupling components **1122** may take the form of any of the examples discussed above with reference to the coupling components **1116**, and the IC package **1124** may take the form of any of the examples discussed above with reference to the IC package **1120**.

[0069] The IC device assembly **1100** illustrated in FIG. **11** includes a package-on-package structure **1134** coupled to the second face **1142** of the circuit board **1102** by coupling components **1128**. The package-on-package structure **1134** may include a first IC package **1126** and a second IC package **1132** coupled together by coupling components **1130** such that the first IC package **1126** is disposed between the circuit board **1102** and the second IC package **1132**. The coupling components **1128**, **1130** may take the form of any of the examples of the coupling components **1116** discussed above, and the IC packages **1126**, **1132** may take the form of any of the examples of the IC package **1120** discussed above. The package-on-package structure **1134** may be configured in accordance with any of the package-on-package structures known in the art.

[0070] FIG. **12** is a block diagram of an example electrical device **1200**. For example, any suitable ones of the components of the electrical device **1200** may include one or more of the device assemblies **1100**, IC devices **900**, or dies **802** disclosed herein. A number of components are illustrated in FIG. **12** as included in the electrical device **1200**, but any one or more of these components may be omitted or duplicated, as suitable for the application. In some examples, some or all of the components included in the electrical device **1200** may be attached to one or more motherboards. In some examples, some or all of these components are fabricated onto a single system-on-a-chip (SoC) die.

[0071] Additionally, in various examples, the electrical device **1200** may not include one or more of the components illustrated in FIG. **12**, but the electrical device **1200** may include interface circuitry for coupling to the one or more components. For example, the electrical device **1200** may not include a display **1206**, but may include display interface circuitry (e.g., a connector and driver circuitry) to which a display **1206** may be coupled. In another set of examples, the electrical device **1200** may not include an audio input device **1218** (e.g., microphone) or an audio output device **1208** (e.g., a speaker, a headset, earbuds, etc.), but may include audio input or output device interface circuitry (e.g., connectors and supporting circuitry) to which an audio input device **1218** or audio output device **1208** may be coupled.

[0072] The electrical device **1200** may include programmable circuitry **1202** (e.g., one or more processing devices). The programmable circuitry **1202** may include one or more digital signal processors (DSPs), application-specific integrated circuits (ASICs), central processing units (CPUs), graphics processing units (GPUs), cryptoprocessors (specialized processors that execute cryptographic algorithms within hardware), server processors, or any other suitable processing devices. The electrical device **1200** may include a memory **1204**, which may itself include one or more memory devices such as volatile memory (e.g., dynamic random access memory (DRAM)), nonvolatile memory (e.g., read-only memory (ROM)), flash memory, solid state memory, and/or a hard drive. In some examples, the memory **1204** may include memory that shares a die with the programmable circuitry **1202**. This memory may be used as cache memory and may include embedded dynamic random access memory (eDRAM) or spin transfer torque magnetic random access memory (STT-MRAM).

[0073] In some examples, the electrical device **1200** may include a communication chip **1212** (e.g., one or more communication chips). For example, the communication chip **1212** may be configured for managing wireless communications for the transfer of data to and from the electrical device **1200**. The term "wireless" and its derivatives may be used to describe circuits, devices, systems, methods, techniques, communications channels, etc., that may communicate data through the use of modulated electromagnetic radiation through a nonsolid medium. The term does not imply that the associated devices do not contain any wires, although in some examples they might not.

[0074] The communication chip **1212** may implement any of a number of wireless standards or protocols, including but not limited to Institute for Electrical and Electronic Engineers (IEEE) standards including Wi-Fi (IEEE 802.11 family), IEEE 802.16 standards (e.g., IEEE 802.16-2005 Amendment), Long-Term Evolution (LTE) project along with any amendments, updates, and/or revisions (e.g., advanced LTE project, ultra mobile broadband (UMB) project (also referred to as "3GPP2"), etc.). IEEE 802.16 compatible Broadband Wireless Access (BWA) networks are generally referred to as WiMAX networks, an acronym that stands for Worldwide Interoperability for Microwave Access, which is a certification mark for products that pass conformity and interoperability tests for the IEEE 802.16 standards. The communication chip **1212** may operate in accordance with a Global System for Mobile Communication (GSM), General Packet Radio Service (GPRS), Universal Mobile Telecommunications System (UMTS), High Speed Packet Access (HSPA), Evolved HSPA (E-HSPA), or LTE network. The communication chip **1212** may operate in accordance with Enhanced Data for GSM Evolution (EDGE), GSM EDGE Radio Access Network (GERAN), Universal Terrestrial Radio Access Network (UTRAN), or Evolved UTRAN (E-UTRAN). The communication chip **1212** may operate in accordance with Code Division Multiple Access (CDMA), Time Division Multiple Access (TDMA), Digital Enhanced Cordless Telecommunications (DECT), Evolution-Data Optimized (EV-DO), and derivatives thereof, as well as any other wireless protocols that are designated as 3G, 4G, 5G, and beyond. The communication chip **1212** may operate in accordance with other wireless protocols in other examples. The electrical device **1200** may include an antenna **1222** to facilitate wireless communications and/or to receive other wireless communications (such as AM or FM radio transmissions).

[0075] In some examples, the communication chip **1212** may manage wired communications, such as electrical, optical, or any other suitable communication protocols (e.g., the Ethernet). As noted above, the communication chip **1212** may include multiple communication chips. For instance, a first communication chip **1212** may be dedicated to shorter-range wireless communications such as Wi-Fi or Bluetooth, and a second communication chip **1212** may be dedicated to longer-range wireless communications such as global positioning system (GPS), EDGE, GPRS, CDMA, WiMAX, LTE, EV-DO, or others. In some examples, a first communication chip **1212** may be dedicated to wireless communications, and a second communication chip **1212** may be dedicated to wired communications.

[0076] The electrical device **1200** may include battery/power circuitry **1214**. The battery/power circuitry **1214** may include one or more energy storage devices (e.g., batteries or capacitors) and/or circuitry for coupling components of the electrical device **1200** to an energy source separate from the electrical device **1200** (e.g., AC line power).

[0077] The electrical device **1200** may include a display **1206** (or corresponding interface circuitry, as discussed above). The display **1206** may include any visual indicators, such as a heads-up display, a computer monitor, a projector, a touchscreen display, a liquid crystal display (LCD), a light-emitting diode display, or a flat panel display.

[0078] The electrical device **1200** may include an audio output device **1208** (or corresponding interface circuitry, as discussed above). The audio output device **1208** may include any device that generates an audible indicator, such as speakers, headsets, or earbuds.

[0079] The electrical device **1200** may include an audio input device **1218** (or corresponding interface circuitry, as discussed above). The audio input device **1218** may include any device that

generates a signal representative of a sound, such as microphones, microphone arrays, or digital instruments (e.g., instruments having a musical instrument digital interface (MIDI) output).
[0080] The electrical device **1200** may include GPS circuitry **1216**. The GPS circuitry **1216** may be in communication with a satellite-based system and may receive a location of the electrical device **1200**, as known in the art.
[0081] The electrical device **1200** may include any other output device **1210** (or corresponding interface circuitry, as discussed above). Examples of the other output device **1210** may include an audio codec, a video codec, a printer, a wired or wireless transmitter for providing information to other devices, or an additional storage device.
[0082] The electrical device **1200** may include any other input device **1220** (or corresponding interface circuitry, as discussed above). Examples of the other input device **1220** may include an accelerometer, a gyroscope, a compass, an image capture device, a keyboard, a cursor control device such as a mouse, a stylus, a touchpad, a bar code reader, a Quick Response (QR) code reader, any sensor, or a radio frequency identification (RFID) reader.
[0083] The electrical device **1200** may have any desired form factor, such as a hand-held or mobile electrical device (e.g., a cell phone, a smart phone, a mobile internet device, a music player, a tablet computer, a laptop computer, a netbook computer, an ultrabook computer, a personal digital assistant (PDA), an ultra mobile personal computer, etc.), a desktop electrical device, a server or other networked computing component, a printer, a scanner, a monitor, a set-top box, an entertainment control unit, a vehicle control unit, a digital camera, a digital video recorder, or a wearable electrical device. In some examples, the electrical device **1200** may be any other electronic device that processes data.
[0084] “Including” and “comprising” (and all forms and tenses thereof) are used herein to be open ended terms. Thus, whenever a claim employs any form of “include” or “comprise” (e.g., comprises, includes, comprising, including, having, etc.) as a preamble or within a claim recitation of any kind, it is to be understood that additional elements, terms, etc., may be present without falling outside the scope of the corresponding claim or recitation. As used herein, when the phrase “at least” is used as the transition term in, for example, a preamble of a claim, it is open-ended in the same manner as the term “comprising” and “including” are open ended. The term “and/or” when used, for example, in a form such as A, B, and/or C refers to any combination or subset of A, B, C such as (1) A alone, (2) B alone, (3) C alone, (4) A with B, (5) A with C, (6) B with C, or (7) A with B and with C. As used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing structures, components, items, objects and/or things, the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. As used herein in the context of describing the performance or execution of processes, instructions, actions, activities, etc., the phrase “at least one of A and B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B. Similarly, as used herein in the context of describing the performance or execution of processes, instructions, actions, activities, etc., the phrase “at least one of A or B” is intended to refer to implementations including any of (1) at least one A, (2) at least one B, or (3) at least one A and at least one B.
[0085] As used herein, singular references (e.g., “a”, “an”, “first”, “second”, etc.) do not exclude a plurality. The term “a” or “an” object, as used herein, refers to one or more of that object. The terms “a” (or “an”), “one or more”, and “at least one” are used interchangeably herein. Furthermore, although individually listed, a plurality of means, elements, or actions may be implemented by, e.g., the same entity or object. Additionally, although individual features may be included in different examples or claims, these may possibly be combined, and the inclusion in different examples or claims does not imply that a combination of features is not feasible and/or

advantageous.
[0086] As used herein, unless otherwise stated, the term “above” describes the relationship of two parts relative to Earth. A first part is above a second part, if the second part has at least one part between Earth and the first part. Likewise, as used herein, a first part is “below” a second part when the first part is closer to the Earth than the second part. As noted above, a first part can be above or below a second part with one or more of: other parts therebetween, without other parts therebetween, with the first and second parts touching, or without the first and second parts being in direct contact with one another.
[0087] Notwithstanding the foregoing, in the case of referencing a semiconductor device (e.g., a transistor), a semiconductor die containing a semiconductor device, and/or an integrated circuit (IC) package containing a semiconductor die during fabrication or manufacturing, “above” is not with reference to Earth, but instead is with reference to an underlying substrate on which relevant components are fabricated, assembled, mounted, supported, or otherwise provided. Thus, as used herein and unless otherwise stated or implied from the context, a first component within a semiconductor die (e.g., a transistor or other semiconductor device) is “above” a second component within the semiconductor die when the first component is farther away from a substrate (e.g., a semiconductor wafer) during fabrication/manufacturing than the second component on which the two components are fabricated or otherwise provided. Similarly, unless otherwise stated or implied from the context, a first component within an IC package (e.g., a semiconductor die) is “above” a second component within the IC package during fabrication when the first component is farther away from a printed circuit board (PCB) to which the IC package is to be mounted or attached. It is to be understood that semiconductor devices are often used in orientation different than their orientation during fabrication. Thus, when referring to a semiconductor device (e.g., a transistor), a semiconductor die containing a semiconductor device, and/or an integrated circuit (IC) package containing a semiconductor die during use, the definition of “above” in the preceding paragraph (i.e., the term “above” describes the relationship of two parts relative to Earth) will likely govern based on the usage context.
[0088] As used in this patent, stating that any part (e.g., a layer, film, area, region, or plate) is in any way on (e.g., positioned on, located on, disposed on, or formed on, etc.) another part, indicates that the referenced part is either in contact with the other part, or that the referenced part is above the other part with one or more intermediate part(s) located therebetween.
[0089] As used herein, connection references (e.g., attached, coupled, connected, and joined) may include intermediate members between the elements referenced by the connection reference and/or relative movement between those elements unless otherwise indicated. As such, connection references do not necessarily infer that two elements are directly connected and/or in fixed relation to each other. As used herein, stating that any part is in “contact” with another part is defined to mean that there is no intermediate part between the two parts.
[0090] Unless specifically stated otherwise, descriptors such as “first,” “second,” “third,” etc., are used herein without imputing or otherwise indicating any meaning of priority, physical order, arrangement in a list, and/or ordering in any way, but are merely used as labels and/or arbitrary names to distinguish elements for ease of understanding the disclosed examples. In some examples, the descriptor “first” may be used to refer to an element in the detailed description, while the same element may be referred to in a claim with a different descriptor such as “second” or “third.” In such instances, it should be understood that such descriptors are used merely for identifying those elements distinctly within the context of the discussion (e.g., within a claim) in which the elements might, for example, otherwise share a same name.
[0091] As used herein, “approximately” and “about” modify their subjects/values to recognize the potential presence of variations that occur in real world applications. For example, “approximately” and “about” may modify dimensions that may not be exact due to manufacturing tolerances and/or other real world imperfections as will be understood by persons of ordinary skill in the art. For

example, “approximately” and “about” may indicate such dimensions may be within a tolerance range of +/−10% unless otherwise specified herein.

[0092] As used herein “substantially real time” refers to occurrence in a near instantaneous manner recognizing there may be real world delays for computing time, transmission, etc. Thus, unless otherwise specified, “substantially real time” refers to real time +1second.

[0093] As used herein, the phrase “in communication,” including variations thereof, encompasses direct communication and/or indirect communication through one or more intermediary components, and does not require direct physical (e.g., wired) communication and/or constant communication, but rather additionally includes selective communication at periodic intervals, scheduled intervals, aperiodic intervals, and/or one-time events.

[0094] As used herein, “programmable circuitry” is defined to include (i) one or more special purpose electrical circuits (e.g., an application specific circuit (ASIC)) structured to perform specific operation(s) and including one or more semiconductor-based logic devices (e.g., electrical hardware implemented by one or more transistors), and/or (ii) one or more general purpose semiconductor-based electrical circuits programmable with instructions to perform specific functions(s) and/or operation(s) and including one or more semiconductor-based logic devices (e.g., electrical hardware implemented by one or more transistors). Examples of programmable circuitry include programmable microprocessors such as Central Processor Units (CPUs) that may execute first instructions to perform one or more operations and/or functions, Field Programmable Gate Arrays (FPGAs) that may be programmed with second instructions to cause configuration and/or structuring of the FPGAs to instantiate one or more operations and/or functions corresponding to the first instructions, Graphics Processor Units (GPUs) that may execute first instructions to perform one or more operations and/or functions, Digital Signal Processors (DSPs) that may execute first instructions to perform one or more operations and/or functions, XPUs, Network Processing Units (NPUs) one or more microcontrollers that may execute first instructions to perform one or more operations and/or functions and/or integrated circuits such as Application Specific Integrated Circuits (ASICs). For example, an XPU may be implemented by a heterogeneous computing system including multiple types of programmable circuitry (e.g., one or more FPGAs, one or more CPUs, one or more GPUs, one or more NPUs, one or more DSPs, etc., and/or any combination(s) thereof), and orchestration technology (e.g., application programming interface(s) (API(s)) that may assign computing task(s) to whichever one(s) of the multiple types of programmable circuitry is/are suited and available to perform the computing task(s).

[0095] As used herein integrated circuit/circuitry is defined as one or more semiconductor packages containing one or more circuit elements such as transistors, capacitors, inductors, resistors, current paths, diodes, etc. For example, an integrated circuit may be implemented as one or more of an ASIC, an FPGA, a chip, a microchip, programmable circuitry, a semiconductor substrate coupling multiple circuit elements, a system on chip (SoC), etc.

[0096] From the foregoing, it will be appreciated that example systems, apparatus, articles of manufacture, and methods have been disclosed that connect a semiconductor package to a circuit board. More particularly, examples disclosed herein resolve challenges of more impedance discontinuities and signal reflections arising from circuit assemblies using PoINT devices having an additional interconnect transition between the dies and/or packages and the interposer. In some examples PoINT devices disclosed herein a first package (e.g., a processor package) is to communicate directly with other components mounted onto the interposer without discontinuities defined by the transitions between the interposer and the socket, and between the socket and the circuit board. Furthermore, in some examples, the first package of some example PoINT devices disclosed herein include interconnects that bypass the interposer to directly connect with an underlying socket on a circuit board to communicate with other components via the circuit board, thereby reducing the number of mid-level interconnect interfaces between the semiconductor packages to the circuit board (e.g., without the mid-level interconnect interface between the package and an interposer).

[0097] Further examples and combinations thereof include the following:

[0098] Example 1 includes an apparatus comprising an interposer having a first side and a second side opposite the first side, and a semiconductor package coupled to the first side of the interposer via first contacts on a first surface of the semiconductor package, the semiconductor package including second contacts on the first surface of the semiconductor package, the second contacts spaced apart from the first contacts, the second contacts to electrical couple to third contacts in a socket dimensioned to receive the interposer and the semiconductor package.

[0099] Example 2 includes the apparatus of example 1, wherein the semiconductor package is a first semiconductor package, the apparatus further including a second semiconductor package coupled to the first side of the interposer via fourth contacts on the second semiconductor package.

[0100] Example 3 includes the apparatus of example 2, wherein the first semiconductor package is electrically coupled to the second semiconductor package via the interposer and independent of the socket, the first semiconductor package to be electrically coupled to an external component via the socket and independent of the interposer.

[0101] Example 4 includes the apparatus of any one of examples 2 or 3, wherein the first semiconductor package is a processor package and the second semiconductor package is a memory package.

[0102] Example 5 includes the apparatus of any one of examples 2-4, wherein the interposer is to be between the fourth contacts of the second semiconductor package and the socket, the interposer is to be between the first contacts of the first semiconductor package and the socket, and the interposer is not to be between the second contacts of the first semiconductor package and the socket.

[0103] Example 6 includes the apparatus of any one of examples 1-5, wherein the first contacts on the semiconductor package have a first shape and the second contacts on the semiconductor package have a second shape, the second shape different from the first shape.

[0104] Example 7 includes the apparatus of any one of examples 1-6, wherein the first contacts on the semiconductor package include BGA pads and the second contacts on the semiconductor package include LGA pads.

[0105] Example 8 includes the apparatus of any one of examples 1-7, wherein the interposer includes a cutout aligned with the second contacts of the semiconductor package so that the interposer is spaced apart from the second contacts.

[0106] Example 9 includes the apparatus of example 8, wherein the socket is to extend into the cutout towards the semiconductor package.

[0107] Example 10 includes the apparatus of any one of examples 1-9, further including a compressible material to be between the interposer and the socket.

[0108] Example 11 includes the apparatus of example 10, wherein the compressible material is attached to the second side of the interposer.

[0109] Example 12 includes an apparatus comprising an interposer to fit into a socket, a first surface of the interposer to face towards the socket, the interposer including an opening extending from the first surface to a second surface of the interposer, the second surface opposite the first surface, and a semiconductor package coupled to the second surface of the interposer, the semiconductor package to extend across the opening of the interposer, the semiconductor package including contacts exposed through the opening in the interposer, the contacts to electrically couple to a portion of the socket that is to extend through the opening in the interposer.

[0110] Example 13 includes the apparatus of example 12, wherein the interposer is devoid of electrical connections with the socket along the second surface of the interposer.

[0111] Example 14 includes the apparatus of any one of examples 12 or 13, further including a peripheral component connector coupled to the second surface of the interposer, the peripheral component connector electrically coupled to the semiconductor package via the interposer and

independent of the socket, the peripheral component connector including at least one of a peripheral component interconnect express (PCIe) connector or a photonic integrated circuit (PIC).
[0112] Example 15 includes an apparatus comprising a socket to be mounted on a circuit board, a substrate to be inserted into the socket, the substrate including a core, the core including woven glass fibers embedded in an epoxy resin, and a semiconductor die to be electrically coupled to the substrate via first contacts and to be electrically coupled to the circuit board via pins in the socket, the substrate to be spaced apart from the pins.
[0113] Example 16 includes the apparatus of example 15, wherein the semiconductor die is a first semiconductor die, the apparatus further including a second semiconductor die, the second semiconductor die electrically coupled to the first semiconductor die through the substrate and independent of the socket.
[0114] Example 17 includes the apparatus of any one of examples 15 or 16, wherein the socket includes a first portion having a first thickness and a second portion having a second thickness, the first thickness less than the second thickness, the first portion positioned between the semiconductor die and the circuit board when the semiconductor die is coupled to the socket, less than all of the second portion positioned between the semiconductor die and the circuit board when the semiconductor die is coupled to the socket.
[0115] Example 18 includes the apparatus of example 17, wherein the first portion is between the substrate and the circuit board when the substrate is inserted into the socket, and no part of the second portion is between the substrate and the circuit board when the substrate is inserted into the socket.
[0116] Example 19 includes the apparatus of any one of examples 17 or 18, wherein the second portion of the socket includes the pins, the pins to extend through the second thickness of the second portion.
[0117] Example 20 includes the apparatus of any one of examples 17-19, wherein electrical connections between the socket and the circuit board are aligned with the second portion of the socket and spaced apart from the first portion of the socket.
[0118] The following claims are hereby incorporated into this Detailed Description by this reference. Although certain example systems, apparatus, articles of manufacture, and methods have been disclosed herein, the scope of coverage of this patent is not limited thereto. On the contrary, this patent covers all systems, apparatus, articles of manufacture, and methods fairly falling within the scope of the claims of this patent.

Claims

1. An apparatus comprising: an interposer having a first side and a second side opposite the first side; and a semiconductor package coupled to the first side of the interposer via first contacts on a first surface of the semiconductor package, the semiconductor package including second contacts on the first surface of the semiconductor package, the second contacts spaced apart from the first contacts, the second contacts to electrical couple to third contacts in a socket dimensioned to receive the interposer and the semiconductor package.
2. The apparatus of claim 1, wherein the semiconductor package is a first semiconductor package, the apparatus further including a second semiconductor package coupled to the first side of the interposer via fourth contacts on the second semiconductor package.
3. The apparatus of claim 2, wherein the first semiconductor package is electrically coupled to the second semiconductor package via the interposer and independent of the socket, the first semiconductor package to be electrically coupled to an external component via the socket and independent of the interposer.
4. The apparatus of claim 2, wherein the first semiconductor package is a processor package and the second semiconductor package is a memory package.

5. The apparatus of claim 2, wherein the interposer is to be between the fourth contacts of the second semiconductor package and the socket, the interposer is to be between the first contacts of the first semiconductor package and the socket, and the interposer is not to be between the second contacts of the first semiconductor package and the socket.
6. The apparatus of claim 1, wherein the first contacts on the semiconductor package have a first shape and the second contacts on the semiconductor package have a second shape, the second shape different from the first shape.
7. The apparatus of claim 1, wherein the first contacts on the semiconductor package include BGA pads and the second contacts on the semiconductor package include LGA pads.
8. The apparatus of claim 1, wherein the interposer includes a cutout aligned with the second contacts of the semiconductor package so that the interposer is spaced apart from the second contacts.
9. The apparatus of claim 8, wherein the socket is to extend into the cutout towards the semiconductor package.
10. The apparatus of claim 1, further including a compressible material to be between the interposer and the socket.
11. The apparatus of claim 10, wherein the compressible material is attached to the second side of the interposer.
12. An apparatus comprising: an interposer to fit into a socket, a first surface of the interposer to face towards the socket, the interposer including an opening extending from the first surface to a second surface of the interposer, the second surface opposite the first surface; and a semiconductor package coupled to the second surface of the interposer, the semiconductor package to extend across the opening of the interposer, the semiconductor package including contacts exposed through the opening in the interposer, the contacts to electrically couple to a portion of the socket that is to extend through the opening in the interposer.
13. The apparatus of claim 12, wherein the interposer is devoid of electrical connections with the socket along the second surface of the interposer.
14. The apparatus of claim 12, further including a peripheral component connector coupled to the second surface of the interposer, the peripheral component connector electrically coupled to the semiconductor package via the interposer and independent of the socket, the peripheral component connector including at least one of a peripheral component interconnect express (PCIe) connector or a photonic integrated circuit (PIC).
15. An apparatus comprising: a socket to be mounted on a circuit board; a substrate to be inserted into the socket, the substrate including a core, the core including woven glass fibers embedded in an epoxy resin; and a semiconductor die to be electrically coupled to the substrate via first contacts and to be electrically coupled to the circuit board via pins in the socket, the substrate to be spaced apart from the pins.
16. The apparatus of claim 15, wherein the semiconductor die is a first semiconductor die, the apparatus further including a second semiconductor die, the second semiconductor die electrically coupled to the first semiconductor die through the substrate and independent of the socket.
17. The apparatus of claim 15, wherein the socket includes a first portion having a first thickness and a second portion having a second thickness, the first thickness less than the second thickness, the first portion positioned between the semiconductor die and the circuit board when the semiconductor die is coupled to the socket, less than all of the second portion positioned between the semiconductor die and the circuit board when the semiconductor die is coupled to the socket.
18. The apparatus of claim 17, wherein the first portion is between the substrate and the circuit board when the substrate is inserted into the socket, and no part of the second portion is between the substrate and the circuit board when the substrate is inserted into the socket.
19. The apparatus of claim 17, wherein the second portion of the socket includes the pins, the pins to extend through the second thickness of the second portion.

20. The apparatus of claim 17, wherein electrical connections between the socket and the circuit board are aligned with the second portion of the socket and spaced apart from the first portion of the socket.